

Notes: Avery & Zemsky (AER, 1998)
Multidimensional Uncertainty and Herd Behavior in Financial Markets

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1 Big picture

- **Question.** Can rational herd behavior occur in financial markets once asset prices adjust to order flow? Can rational herding create bubbles, crashes, or excess volatility?
- **Core answer.** With only one dimension of uncertainty, competitive price adjustment eliminates herding. With multiple dimensions of uncertainty, a one-dimensional price cannot summarize everything traders know, so rational history-dependent trading can arise.
- **One-dimensional benchmark.** When private signals are monotonic and concern only the asset value, informed traders always trade according to their signal. Price adjustment prevents fixed-price informational cascades.
- **Event uncertainty.** If traders are uncertain whether an information event has occurred, informed traders may trade with the trend even when their value signal is weak or contrary.
- **Composition uncertainty.** If the market is also uncertain about how precise traders are, herd behavior can cause substantial short-run mispricing.
- **Economic takeaway.** Rational prices rule out long-run cascades, but can be temporarily overwhelmed when order flow communicates several kinds of information at once.

2 Incremental contribution of the paper

- **Endogenizes price in herding models.** Earlier cascade models largely fixed the price of taking an action. Avery and Zemsky embed sequential learning in a Glosten–Milgrom style market where bid and ask prices respond rationally to trade history.
- **Separates cascades from herds.** The paper shows that financial markets can rule out true informational cascades while still permitting rational herd behavior.
- **Identifies a dimensionality logic.** Herding and bubbles require more dimensions of uncertainty than the price can reveal at once.
- **Explains rational bubbles without irrationality.** The simulation demonstrates a bubble generated by Bayesian traders and a competitive market maker.
- **Rationalizes charting.** With multidimensional uncertainty, histories of trades and prices can carry information not summarized by the current price alone.

3 Model primitives and measurement

3.1 Sequential trading environment

There is one risky asset with true value $V \in [0, 1]$. A competitive risk-neutral market maker interacts with an infinite sequence of traders. The public history before period t is H_t . Each arriving trader can buy one unit, sell one unit, or not trade. The market maker sets bid and ask prices to break even:

$$B_t = E[V \mid h_t = S, H_t], \quad A_t = E[V \mid h_t = B, H_t].$$

Interpretation. The bid and ask are Bayesian conditional expectations that include adverse-selection information from the incoming order.

3.2 Trader types and signals

With probability μ an informed trader arrives; with probability $1 - \mu$ a noise trader arrives. Noise traders buy, sell, or do not trade with equal probability. An informed trader of type θ observes signal x_θ drawn from $f_\theta(x | V)$. The informed trader's valuation is

$$V_t(x_\theta) = E[V | H_t, x_\theta], \quad V_t = E[V | H_t].$$

Interpretation. Herd behavior is defined by comparing a trader's initial signal-based valuation with his action after observing market history.

3.3 Bayesian updating and cascades

$$\pi_{t+1}^v = P(V = v | H_t, h_t) = \frac{\pi_t^v P(h_t | V = v, H_t)}{P(h_t, H_t)}.$$

An informational cascade occurs when

$$P(h_t | V, H_t) = P(h_t | H_t) \quad \text{for all } V, h_t.$$

Interpretation. A cascade means actions stop revealing information. The paper shows this cannot happen in equilibrium with the price mechanism and noise trading.

4 Models and theoretical specifications

4.1 Fixed-price cascade benchmark with endogenous price

In the Bikhchandani–Hirshleifer–Welch benchmark, a fixed price can make later agents ignore private signals. Avery and Zemsky replace the fixed cost with an endogenous price $c_t = E[V | H_t]$. With binary V , $c_t = \pi_t = P(V = 1 | H_t)$.

Interpretation. When price moves with public beliefs, it neutralizes the informational advantage of simple imitation.

4.2 Propositions 1–4: price discipline and learning

- Proposition 1: unique zero-profit bid and ask prices exist, satisfy $B_t \leq V_t \leq A_t$, and beliefs are martingales.
- Proposition 2: informational cascades never occur in market equilibrium.
- Proposition 3: monotonic signals rule out herd behavior.
- Proposition 4: bid and ask prices converge almost surely to the true asset value.

Interpretation. These results establish the benchmark: with one-dimensional value uncertainty, rational pricing prevents herd-driven inefficiency.

4.3 Event uncertainty

Event uncertainty means that the market is unsure whether an information event has occurred. Informed traders may learn both that an event occurred and whether it was good or bad. Proposition 5 shows that herding can occur with positive probability when signals are imperfect. Proposition 7 shows that herding can become prolonged when event uncertainty is extreme.

Interpretation. Event uncertainty creates herding because trades reveal both event occurrence and value direction. But prices need not become severely distorted because the market maker discounts herd trades.

4.4 Composition uncertainty

Composition uncertainty means the market is unsure about the precision of informed traders. A well-informed market has many high-quality traders; a poorly informed market has few. If the market initially expects a well-informed market, a sequence of herd buys can be misread as high-quality information, producing a bubble.

Interpretation. The bubble is not irrational. It is rational Bayesian learning about the wrong dimension of uncertainty.

4.5 Contrarian behavior

Composition uncertainty without event uncertainty can also produce contrarian behavior: traders may trade against the trend because the history reveals that others are better informed.

Interpretation. The result provides a rational reason why traders may care about detailed price histories, not just current prices.

5 Identification and theoretical strategy

- **Start from fixed-price cascades.** The paper shows exactly how endogenous pricing changes cascade logic.
- **Isolate one dimension of uncertainty.** Monotonic value signals provide the no-herding benchmark.
- **Add uncertainty dimensions sequentially.** Event uncertainty generates herding; composition uncertainty makes herding distort prices.
- **Distinguish short-run from long-run.** Martingale and convergence results imply eventual learning, while simulations show short-run bubbles.
- **Use extreme cases to clarify mechanisms.** The price-bubble simulation uses extreme event uncertainty and mistaken composition priors to make the mechanism visible.

Interpretation. The identification is theoretical: it isolates which informational dimensions are necessary for herding and price distortion.

6 Tables and figures

6.1 Figure 1: An example of a price bubble

Read: The simulated price stays near $1/2$, then jumps toward one after early buys induce herd buying, and finally returns toward $1/2$ once the market learns that the economy is poorly informed.

Economic message. The figure shows a rational bubble: a large short-run price movement caused by Bayesian learning about the wrong dimension of uncertainty.

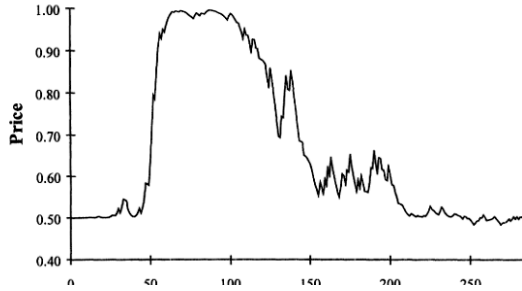


Figure 1: An example of a price bubble



Figure 2: Trading activity, moving average

6.2 Figure 2: Trading activity (moving average)

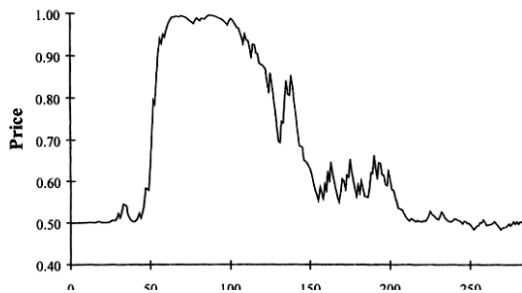
Read: The moving average of buys rises during the herd. Later, no-trade probability rises as spreads push low-quality traders out of the market, helping reveal that earlier buys were not high-quality information.

Economic message. Trading intensity helps diagnose composition. No-trade is informative because it reveals how traders respond to wider spreads.

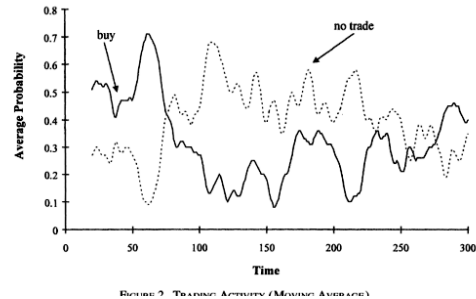
6.3 Figure 1 + Figure 2 together: price path and order-flow diagnosis

Read: Figure 1 shows the price overshoot; Figure 2 shows how buy and no-trade activity explain the later correction.

Economic message. Prices and order flow jointly reveal multidimensional information. The current price alone need not summarize everything during the transition.



(a) Price bubble



(b) Trading activity

7 Interpretive synthesis

- Price adjustment disciplines information aggregation and rules out true cascades.
- Herding is possible when trades simultaneously reveal event occurrence and value direction.
- Bubbles require an additional composition dimension: the market misreads herd trades as evidence from well-informed traders.
- Long-run efficiency and short-run instability can coexist.

8 Short summary

- The paper studies rational herding in financial markets with endogenous bid and ask prices.
- With one-dimensional value uncertainty and monotonic signals, herding is impossible.
- With event uncertainty, rational herding can occur but need not severely distort prices.
- With event plus composition uncertainty, herding can create rational short-run bubbles and crashes.
- The simulation shows a price bubble and the trading activity that eventually resolves it.

9 Conclusion

9.1 Core conclusion

Avery and Zemsky show that rational price adjustment rules out complete informational cascades and eliminates herding when private information is one-dimensional. Multidimensional uncertainty reopens the door to herding and short-run mispricing.

9.2 Economic intuition

A current price is a compact statistic. If order flow only needs to reveal asset value, the price works well. If order flow must reveal event occurrence, value direction, and trader quality, a single price can temporarily confound these meanings.

9.3 Takeaway

The paper shows that rational Bayesian markets can still exhibit herding, charting, bubbles, and crashes when uncertainty has more dimensions than the price can immediately summarize.